

Software Requirements Specification (SRS)

Project: Online Internship & Placement Preparation Portal

Team: Team 12

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Online Internship & Placement Preparation Portal. It defines the functional and non-functional requirements, interfaces, and verification criteria for the system, and serves as a reference for developers, evaluators, and stakeholders.

1.2 Scope

The portal enables students to discover and apply for internships, prepare through aptitude tests, and track their application progress. Recruiters can post internship openings and manage applicants, while admins verify recruiters, manage listings, and monitor placement analytics. The system covers internship listings, applications, aptitude tests, progress tracking, and analytics. It excludes payroll/stipend disbursement and does not replace the college's official placement cell records.

1.3 Audience

Developers, QA Engineers, Students, Recruiters, Placement Coordinators, and Assessment Evaluators.

1.4 Definitions

List of acronyms: SRS (Software Requirements Specification), UI (User Interface), API (Application Programming Interface), JWT (JSON Web Token), OTP (One-Time Password), TLS (Transport Layer Security), NFR (Non-Functional Requirement), RTM (Requirements Traceability Matrix), MERN (MongoDB, Express, React, Node.js).

2. Overall Description

2.1 Product Perspective

The Online Internship & Placement Preparation Portal is a web-based application built on the MERN stack (MongoDB, Express.js, React, Node.js) with Python used for auxiliary analytics/scoring scripts. It consists of a React front-end, a Node/Express REST API back-end, a MongoDB database, and integrated notification (email) services.

2.2 Major Product Functions

- Student and recruiter registration & authentication
- Internship listing creation, search, and filtering
- Internship application submission and status tracking
- Timed aptitude tests with auto-grading
- Student progress dashboard and skill-gap indicators
- Placement analytics and reporting for admins
- Recruiter verification and account/listing administration
- Email/in-app notifications for key events

2.3 User Roles and Characteristics

- Student: basic computer literacy, expects a simple search-apply-track flow and clear test instructions.

- Recruiter: represents a hiring company; needs an efficient way to post roles and review applicants.
- Admin/Placement Coordinator: verifies recruiters, moderates listings, and monitors overall placement metrics.

2.4 Operating Environment

Modern web browsers (Chrome, Firefox, Edge, Safari) on desktop and mobile; Node.js/Express backend deployed on a cloud server; MongoDB Atlas (or equivalent) as the database; standard internet connectivity for client access.

2.5 Constraints

Must comply with basic data-protection norms for storing student PII and resumes; limited to intra-college/placement-cell use in the initial release; dependent on third-party email service for OTP/notifications; developed within an academic project timeline using the MERN stack and Python for supporting scripts.

3. External Interface Requirements

3.1 User Interfaces

A responsive React-based web UI with dashboards for Student, Recruiter, and Admin roles. Forms include clear validation messages; test-taking UI shows a persistent countdown timer.

3.2 Hardware Interfaces

None beyond standard client devices (desktop/laptop/mobile) with a keyboard, mouse/touch input, and display capable of rendering a modern web browser.

3.3 Software Interfaces

- REST API (Node/Express) over HTTPS for all client-server communication.
- MongoDB for persistent storage of users, listings, applications, and test data.
- Email service API (e.g., SMTP/SendGrid) for OTP and status notifications.
- Python-based scoring/analytics scripts invoked by the backend for aptitude test grading and report generation.

3.4 Communications

TLS 1.2+ enforced for all client-server and third-party service communication; JWT tokens used for stateless session authentication; retry logic for transient email-service failures.

4. System Features (Detailed)

Each requirement below includes priority, source, and acceptance criteria / test case reference. IDs follow IPP-F-####.

4.1 User Registration & Authentication

Registration, login, password recovery, and account security for students and recruiters.

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance Criteria / Test Ref
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Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance Criteria / Test Ref
IPP-F-001	The system shall allow students to register using their SRN, college email, and password, with email verification.	Functional	High	Student	AC-001: Verification email sent and account activated only after confirmation. Test: TC-Auth-01
IPP-F-002	The system shall allow recruiters/companies to register and remain in a pending state until verified by an admin.	Functional	High	Recruiter/Admin	AC-002: Unverified recruiters cannot post listings. Test: TC-Auth-02
IPP-F-003	The system shall authenticate registered users via secure login using JWT-based sessions.	Functional	High	All Users	AC-003: Valid credentials return a signed session token. Test: TC-Auth-03
IPP-F-004	The system shall allow users to reset a forgotten password using an OTP sent to their registered email.	Functional	Medium	All Users	AC-004: OTP expires in 10 minutes and is single-use. Test: TC-Auth-04
IPP-F-005	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts.	Functional	High	Security	AC-005: 6th attempt is rejected with lockout message. Test: TC-Auth-05

4.2 Internship Listings & Search

Posting, browsing, filtering, and viewing internship opportunities.

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance Criteria / Test Ref
IPP-F-006	The system shall allow verified recruiters to post internship listings including title, description, required skills, stipend, duration, and location.	Functional	High	Recruiter	AC-006: New listing appears in search within 1 minute of posting. Test: TC-List-01
IPP-F-007	The system shall allow students to search and filter internships by skill, location, stipend, and duration.	Functional	High	Student	AC-007: Filtered results match all applied filter criteria. Test: TC-List-02
IPP-F-008	The system shall display full internship details including eligibility criteria and application deadline.	Functional	Medium	Student	AC-008: All listing fields render correctly on the details page. Test: TC-List-03
IPP-F-009	The system shall notify students of newly posted internships matching their saved skill preferences.	Functional	Medium	Student	AC-009: Notification generated within 5 minutes of a matching post. Test: TC-List-04

4.3 Application Management

Submitting applications and tracking their status through the hiring pipeline.

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance Criteria / Test Ref
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Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance Criteria / Test Ref
IPP-F-010	The system shall allow students to apply to an internship by submitting a resume and optional cover letter.	Functional	High	Student	AC-010: Application record created with status 'Applied'. Test: TC-App-01
IPP-F-011	The system shall let students track application status (Applied, Shortlisted, Rejected, Selected) in real time.	Functional	High	Student	AC-011: Status shown on dashboard matches recruiter's latest update. Test: TC-App-02
IPP-F-012	The system shall allow recruiters to review submitted applications and update their status.	Functional	High	Recruiter	AC-012: Status change is persisted and time-stamped. Test: TC-App-03
IPP-F-013	The system shall send an email/in-app notification to the student whenever their application status changes.	Functional	Medium	Student	AC-013: Notification delivered within 2 minutes of status update. Test: TC-App-04

4.4 Aptitude Test Module

Timed, auto-graded practice tests to help students prepare for placements.

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance Criteria / Test Ref
IPP-F-014	The system shall provide timed aptitude tests categorized by topic (quantitative, logical reasoning, verbal ability).	Functional	High	Student	AC-014: Test auto-submits when the timer reaches zero. Test: TC-Test-01
IPP-F-015	The system shall auto-grade objective-type test questions and display the score immediately after submission.	Functional	High	Student	AC-015: Score displayed matches the correct-answer count. Test: TC-Test-02
IPP-F-016	The system shall record each student's test attempt history, including score and time taken.	Functional	Medium	Student	AC-016: Attempt history visible under student profile. Test: TC-Test-03

4.5 Progress Tracking & Analytics

Student-facing dashboards and admin-facing placement analytics.

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance Criteria / Test Ref
IPP-F-017	The system shall display a student dashboard summarizing applications, test scores, and a basic skill-gap indicator.	Functional	Medium	Student	AC-017: Dashboard widgets reflect the latest data on load. Test: TC-Ana-01
IPP-F-018	The system shall generate placement analytics for admins, including application volume, offers made, and top hiring companies.	Functional	Medium	Admin	AC-018: Report figures match underlying database counts. Test: TC-Ana-02

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance Criteria / Test Ref
IPP-F-019	The system shall recommend relevant internships to a student based on aptitude test performance and profile skills.	Functional	Low	Student	AC-019: Recommended list contains only skill-matching internships. Test: TC-Ana-03

4.6 Admin & Recruiter Panel

Administrative controls for recruiter verification and listing/user management.

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance Criteria / Test Ref
IPP-F-020	The system shall allow an admin to approve or reject pending recruiter registration requests.	Functional	High	Admin	AC-020: Rejected recruiters cannot log in or post listings. Test: TC-Admin-01
IPP-F-021	The system shall allow an admin to add, edit, or deactivate internship listings and user accounts.	Functional	Medium	Admin	AC-021: Deactivated listings no longer appear in student search. Test: TC-Admin-02

5. Non-Functional Requirements

NFRs below are measurable and tied to test plans. IDs follow IPP-NF-###.

Req ID	Requirement	Category	Priority	Acceptance / Measurement
IPP-NF-001	Search and page-load responses shall complete within 3 seconds for 90% of requests under normal load.	Performance	High	90th percentile $\leq$ 3s in load test. Test: TC-Perf-01
IPP-NF-002	The system shall provide 99.5% monthly availability, excluding scheduled maintenance windows.	Reliability	High	Uptime reports show $\geq$ 99.5%/month. Test: Ops report
IPP-NF-003	All user passwords shall be stored using salted bcrypt hashing; plaintext storage is prohibited.	Security	High	Password DB audit shows hashed values only. Test: TC-Sec-01
IPP-NF-004	The system shall support at least 5,000 concurrent users without noticeable performance degradation.	Scalability	Medium	Load test with 5,000 virtual users passes SLA. Test: TC-Perf-02
IPP-NF-005	The web UI shall be responsive across desktop and mobile browsers and follow WCAG 2.1 AA basics.	Usability/Accessibility	Medium	Accessibility audit checklist pass. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

- Protect student and recruiter personal data (resumes, contact details, credentials) from unauthorized access or disclosure.

- Ensure the integrity and confidentiality of aptitude test content to prevent leakage or cheating.
- Ensure only verified recruiters can post listings and access student application data.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Test Ref
IPP-SR-001	TLS 1.2+ shall be mandatory for all client-server communication.	Security	High	TC-Sec-02
IPP-SR-002	Passwords shall be hashed with bcrypt and salt; never stored or transmitted in plaintext.	Security	High	TC-Sec-01
IPP-SR-003	Role-based access control (Student, Recruiter, Admin) shall be enforced on every API endpoint.	Security	High	TC-Sec-03
IPP-SR-004	Aptitude test content shall be served via secure, time-bound tokens to prevent reuse or leakage.	Security	High	TC-Sec-04
IPP-SR-005	All authentication and admin actions shall be logged and retained for at least 1 year for audit.	Security	Medium	TC-Sec-05

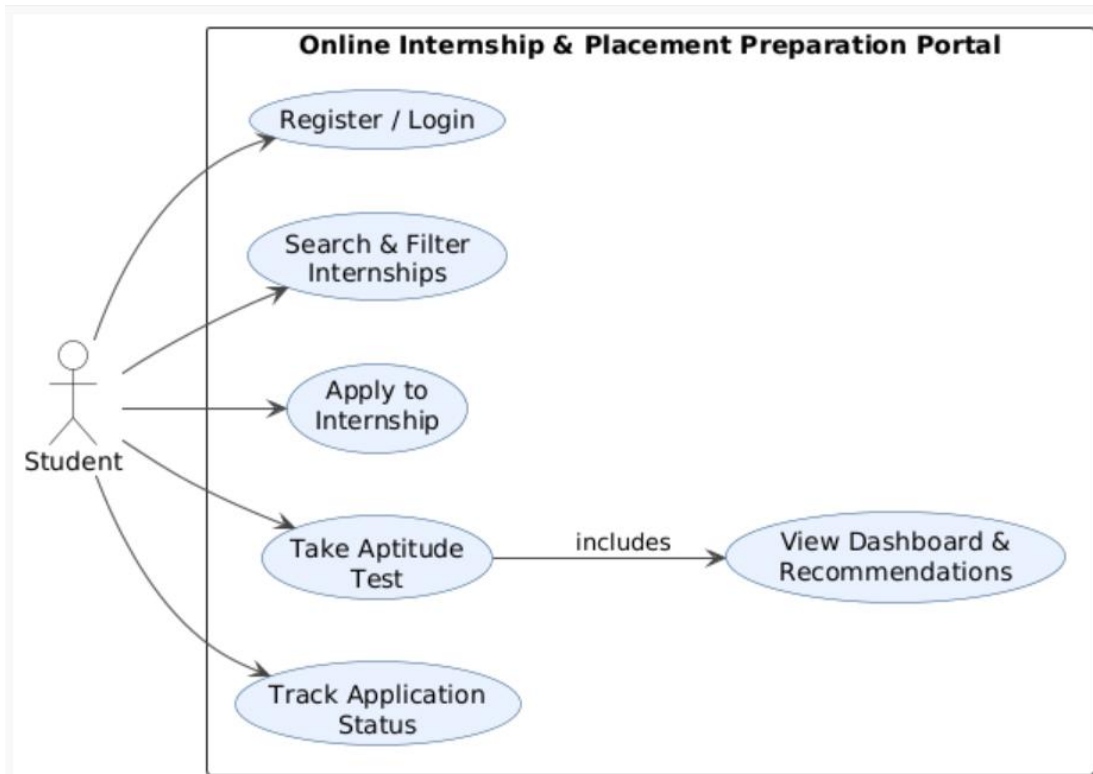
6. Quality Attributes & Acceptance Tests

Exit criteria for acceptance: all High-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all listed test cases passed.

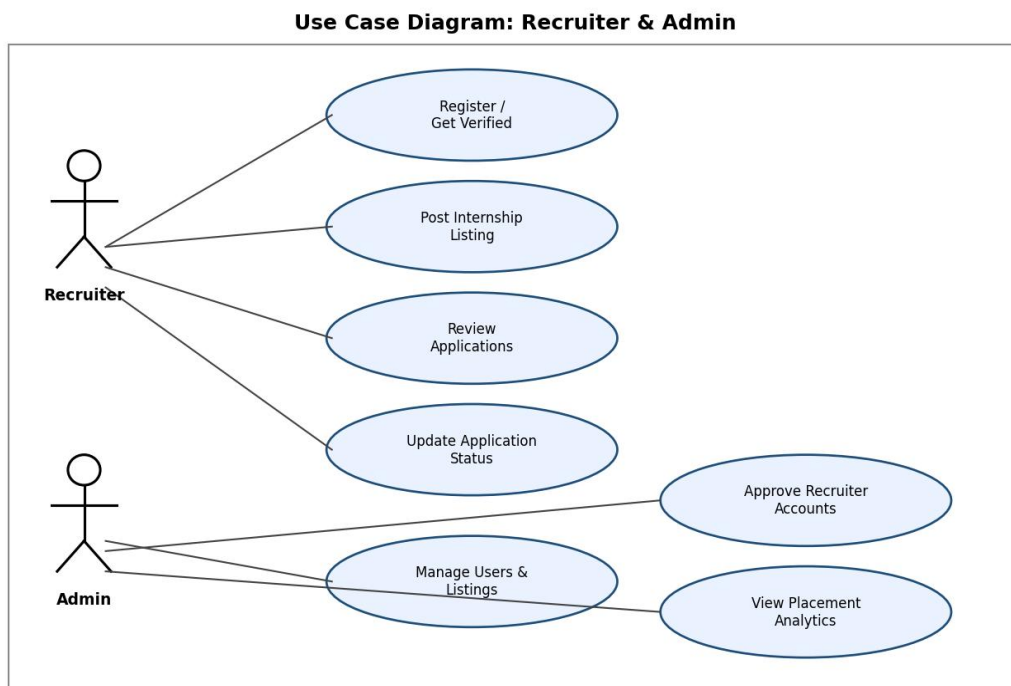
Acceptance test suites: Authentication, Listings & Search, Applications, Aptitude Tests, Analytics, Security, and Usability/Accessibility.

7. System Models and Diagrams

7.1 UML Use-Case Diagram — Student



7.2 UML Use-Case Diagram — Recruiter & Admin



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement (short)	Section Ref	Module	Test Case(s)	Status	Comments
IPP-F-001	Student registration & verification	4.1	AuthModule	TC-Auth-01	N	
IPP-F-003	Secure login (JWT)	4.1	AuthModule	TC-Auth-03	N	
IPP-F-006	Post internship listing	4.2	ListingModule	TC-List-01	N	
IPP-F-007	Search & filter internships	4.2	ListingModule	TC-List-02	N	
IPP-F-010	Submit application	4.3	ApplicationModule	TC-App-01	N	
IPP-F-011	Track application status	4.3	ApplicationModule	TC-App-02	N	
IPP-F-014	Timed aptitude test	4.4	TestModule	TC-Test-01	N	
IPP-F-015	Auto-grade test	4.4	TestModule	TC-Test-02	N	
IPP-F-018	Placement analytics report	4.5	AnalyticsModule	TC-Ana-02	N	
IPP-F-020	Approve recruiter accounts	4.6	AdminModule	TC-Admin-01	N	
IPP-NF-001	Response time target	5	WebUI / API	TC-Perf-01	N	
IPP-SR-002	Password hashing	5.1.2	AuthModule	TC-Sec-01	N	